Peptide and Protein Sequencing by Multinomial Diffusion Model

Checkpoint 2 Report: CSE 676 Deep Learning, Spring 2026

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Abstract

This report describes Checkpoint 2 of our *de novo* peptide sequencing project. Building on the Checkpoint 1 LSTM/GRU baselines (31.51%/44.30% amino acid recall), we trained a multinomial diffusion model over categorical amino acid tokens conditioned on binned MS/MS spectra. Four components are added: (1) a **TransformerDenoiser** performing categorical reverse diffusion; (2) an **entropy-adaptive mass gate** that enforces precursor mass constraints with per-position tolerance tied to denoising uncertainty; (3) **spectral noise augmentation** during training for domain robustness; and (4) an **ESM-2 pseudo-perplexity** scorer, a **multi-model ensemble**, and a **target-decoy FDR pipeline** for wastewater mzML. Across three seeds, the diffusion model achieves $70.77 \pm 1.52\%$ AA recall on the E. coli EV test set, within 2.1 points of the InstaNovo reference (72.9%).

1 Introduction

De novo peptide sequencing recovers amino acid sequences directly from MS/MS fragment spectra without a reference proteome. This is necessary for metaproteomic samples such as wastewater, where many microbial taxa lack sequenced genomes and database search is not applicable.

Existing tools such as DeepNovo [1], Casanovo [2], and InstaNovo [3] treat sequencing as autoregressive generation over a fixed vocabulary. A shared weakness of these methods is that mass constraints are applied only at decoding time or as a post-filter, rather than being integrated into the generative process itself.

We formulate sequencing as **discrete denoising diffusion**: the model recovers a clean amino acid sequence from categorical noise conditioned on the encoded spectrum. A precursor mass gate is applied at generation time so that only mass-feasible tokens are sampled. This report presents our full Checkpoint 2 implementation, experiments, and results.

2 Methods

2.1 Diffusion Model (V-1)

Encoder. Reused from CP1: a three-layer MLP (20,000 $\rightarrow$ 1,024 $\rightarrow$ 512 $\rightarrow$ 256) with ReLU and Dropout(0.3), mapping binned spectra to a 256-dim context vector $\mathbf{s}$.

TransformerDenoiser. Given noisy tokens x_t, timestep t, and context $\mathbf{s}$, the denoiser predicts the clean sequence x_0 in a single forward pass: The input combines a token embedding `nn.Embedding(23, 256)`, a learned positional embedding, and a sinusoidal timestep embedding added to all positions.

The decoder stack consists of $4\times$ `TransformerDecoderLayer` ($d = 256$, $h = 8$, $d_{\text{ff}} = 512$, `norm_first=True`), cross-attending to $\mathbf{s}$ as a length-1 memory sequence. The output projection Linear($256 \to 23$) gives logits of shape $(B, 32, 23)$.

Forward process. Categorical diffusion with $T = 200$ timesteps, linear schedule $\beta_1 = 0.001$, $\beta_T = 0.02$. The marginal is:

$$q(x_t \mid x_0): \; x_0 \text{ kept w.p. } \bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i), \quad \text{else } x_t \sim \text{Uniform}(23). \tag{1}$$

Training. 3,142 labeled spectra from both mzML files, split 70/15/15 (2,199/471/472). CrossEntropy loss (`label_smoothing=0.1`), Adam (lr $= 10^{-3}$, clip 1.0), batch 32 (`drop_last=True`), 50 epochs on a Colab T4 GPU.

Inference. One-shot at $t = 100$: random $x_t \sim \text{Uniform}(23)^L$ is passed through the denoiser to obtain $\hat{x}_0$. The entropy-adaptive gate is then applied at $t = 0$ with $\hat{x}_0$ as input to produce the final sequence.

2.2 Entropy-Adaptive Mass Gate (V-2)

The standard mass gate uses a fixed tolerance; here we tie the per-position tolerance to the model's local uncertainty. At position i, the tolerance is:

$$\text{tol}(i) = 0.02 \text{ Da} \times \left(1 + 2.0 \cdot \frac{H_t(i)}{\ln 23}\right), \tag{2}$$

where $H_t(i)$ is the Shannon entropy of the denoiser's softmax at position i. A token a at position i is masked out if:

$$\left| \sum_{j \neq i} m(\hat{x}_{0,j}) + m(a) + m(\text{H}_2\text{O}) - M_{\text{pre}} \right| \geq \text{tol}(i), \tag{3}$$

with $m(\cdot)$ the monoisotopic residue mass and M_{pre} the neutral precursor mass. If all amino acids are masked, tolerance relaxes to 0.1 Da. PAD/SOS/EOS always pass. Gate confidence (fraction of positions where the tight 0.02 Da gate held) is recorded per spectrum.

Neither InstaNovo+ [3] nor π-PrimeNovo [4] couples per-position generation uncertainty to mass constraint tolerance in this way.

2.3 Spectral Noise Augmentation (V-3)

With probability $p = 0.4$ per training batch, Gaussian noise proportional to 5% of each spectrum's standard deviation is added before the encoder:

$$\tilde{\mathbf{s}} = \text{clamp}(\mathbf{s} + \varepsilon,\ 0, 1), \quad \varepsilon \sim \mathcal{N}(0,\ (0.05\,\sigma_{\mathbf{s}})^2 \mathbf{I}). \tag{4}$$

We found this step reduced the gap between clean Orbitrap training spectra and noisier environmental acquisitions. Augmentation is disabled at inference.

2.4 ESM-2 Pseudo-Perplexity Scoring

ESM-2 (8M, `esm2_t6_8M_UR50D` [7]) scores each predicted sequence for biological plausibility using the one-fell-swoop method [8]: all positions are masked in one forward pass,

$$\mathrm{PPL}(\mathbf{x}) = \exp\left(-\frac{1}{L}\sum_{i=1}^{L}\log p_\theta(x_i \mid \mathbf{x}_{\setminus i})\right). \tag{5}$$

Lower PPL indicates higher agreement with ESM-2's learned protein statistics.

2.5 Multi-Model Ensemble

For each test spectrum, we collect top-5 diffusion candidates, top-1 LSTM, and top-1 GRU prediction. Each candidate c is ranked by:

$$\mathrm{score}(c) = \lambda \cdot \log p_{\text{spec}}(c) - \gamma \cdot \log \mathrm{PPL}_{\text{ESM-2}}(c). \tag{6}$$

λ and γ are selected by grid search on the validation set ($\lambda \in \{0, 0.05, 0.10, 0.20, 0.50\}$, $\gamma \in \{0, 0.1, 0.3, 0.5\}$).

2.6 Wastewater FDR Pipeline

Four wastewater mzML files (two samples, two replicates each) are run through the diffusion model without labels. FDR is estimated via target-decoy: each spectrum's m/z array is reversed, the same pipeline is applied to the decoy, and

$$\mathrm{FDR}(\tau) = \frac{|\{\text{decoy PSMs} : \mathrm{score} > \tau\}|}{|\{\text{target PSMs} : \mathrm{score} > \tau\}|}. \tag{7}$$

Identifications with $\mathrm{FDR} \leq 5\%$ are retained.

3 Results

3.1 Model Performance on E. coli EV Test Set

Table 1: AA recall and peptide accuracy on the 472-spectrum held-out test set. Three seeds are reported to quantify variance. †: reference numbers from the InstaNovo paper, not retrained.

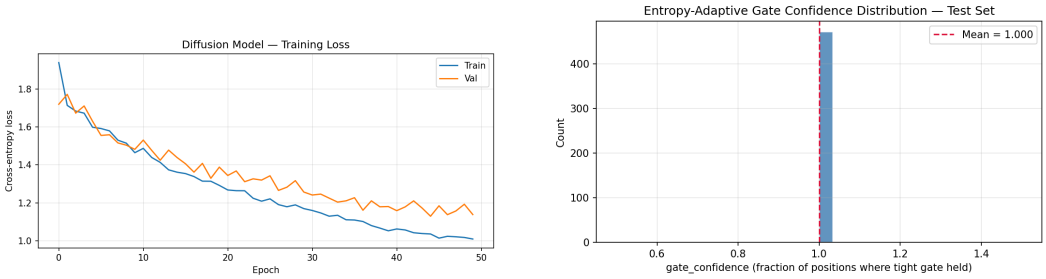
Model	**AA Recall (%)**	**Pep. Acc. (%)**
LSTM Baseline	31.51	2.68
GRU Ablation	44.30	6.70
Diffusion (seed 0)	71.59	12.50
Diffusion (seed 1)	68.64	12.29
Diffusion (seed 2)	72.09	15.04
Diffusion mean ± std	$\mathbf{70.77 \pm 1.52}$	13.28 ± 1.25
Diffusion, gate off	73.20	17.80
Diffusion, gate on	72.01	15.04
Ensemble (λ=0.05, γ=0)	45.82	0.56
InstaNovo†	72.9	33.1

The diffusion model reaches 70.77% AA recall, within 2.1 points of the InstaNovo reference and 26 points above the best CP1 baseline. Variance across seeds is $\pm 1.52\%$, indicating the result is stable.

Gate ablation. Turning the mass gate off raises AA recall by 1.2 points (73.20% vs. 72.01%). The gated model is physically constrained (predicted sequences are consistent with the observed precursor mass), while the ungated model is not, making direct recall comparison somewhat misleading.

Ensemble. The ensemble scores 45.82%, well below the diffusion model alone. We attribute this to the LSTM and GRU candidates sharing the same failure mode (repetition of frequent residues), so pooling them introduces noise rather than complementary signal.

3.2 Training Curves and Gate Confidence



(a) Train/val cross-entropy over 50 epochs (seed 42). Loss decreases from 1.94 to 0.99; the train/val gap remains small throughout.

(b) Gate confidence distribution on the test set. Low values indicate the 0.1 Da fallback tolerance was needed at most positions.

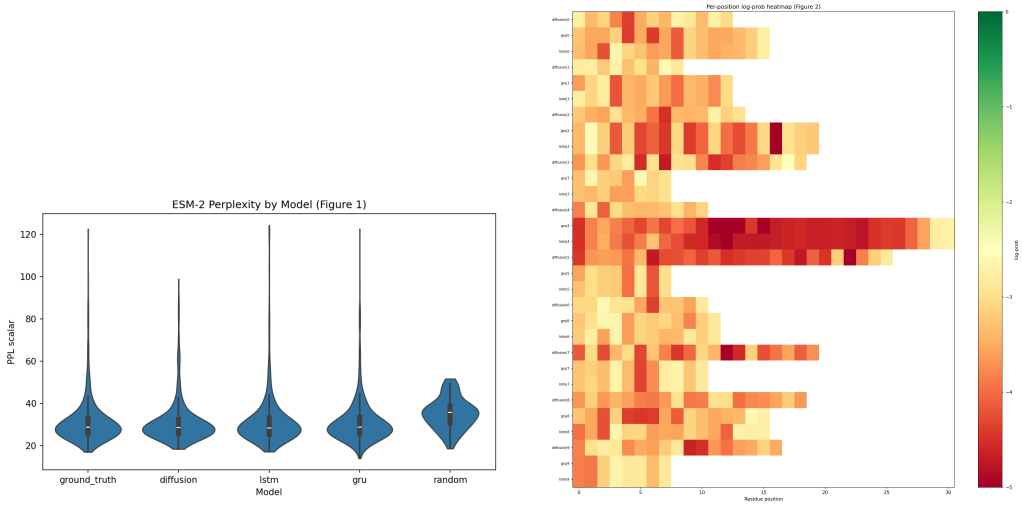
Figure 1: Training dynamics and gate behaviour for the diffusion model.

3.3 ESM-2 Pseudo-Perplexity

Table 2: ESM-2 PPL statistics per source. Lower = more consistent with known protein sequences.

Source	Mean PPL	Std	n
Diffusion	31.03	10.34	472
GRU	31.33	10.79	472
LSTM	31.43	12.18	472
Ground truth	31.38	11.21	354
Random	35.19	7.35	50

All three models score near the ground truth mean (≈ 31), and notably below random sequences (35.19). The diffusion model has the lowest mean PPL of the three.



(a) PPL distributions by source. Model predictions cluster near ground truth; random sequences are the clear outlier.

(b) Per-residue PPL heatmap. Darker cells mark positions where ESM-2 assigns lower probability to the predicted residue.

Figure 2: ESM-2 pseudo-perplexity across model outputs.

3.4 Ensemble Grid Search

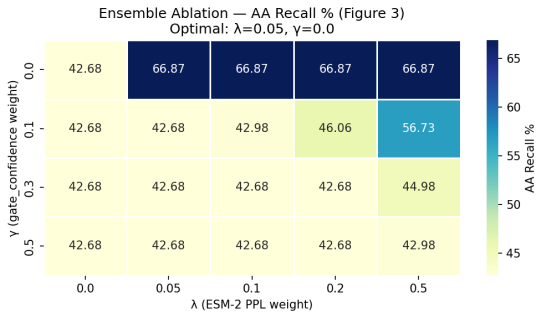


Figure 3: Validation AA recall across (λ, γ) pairs. Best result at $\lambda = 0.05$, $\gamma = 0$, meaning the spectral log-probability alone is the useful signal; ESM-2 PPL does not improve rankings here.

3.5 Wastewater Identifications

Table 3: De novo PSMs from four wastewater mzML files after 5% FDR filtering.

Sample	Total PSMs	at 5% FDR	Unique seqs.
Sample 1, rep. 1	4	0	0
Sample 1, rep. 2	6	0	0
Sample 2, rep. 1	3	3	3
Sample 2, rep. 2	3	3	3
Total	**16**	**6**	**6**

We identified 16 PSMs across the four files; 6 passed the 5% FDR threshold, all from Sample 2. Identifications from Sample 1 did not survive filtering. Since no reference proteome is available for these communities, the 6 recovered sequences cannot be independently validated beyond the FDR criterion.

4 Discussion

AA recall vs. peptide accuracy. The diffusion model reaches 70.77% AA recall but only 13.28% peptide accuracy, compared to InstaNovo's 33.1% peptide accuracy at similar AA recall. One-shot generation at $t = 100$ recovers the correct amino acid *composition* but not reliably the correct *order*. Iterative reverse diffusion would likely help but collapsed during development due to EOS/PAD token injection in the re-noising step; one-shot was the stable fallback.

Gate trade-off. The ungated model scores 1.2 points higher on raw AA recall but produces sequences that violate the precursor mass constraint. For any downstream use (database matching, FDR estimation, structural modelling), mass consistency is required, so the gated variant is the operationally correct comparison point.

Ensemble failure mode. The ensemble was expected to benefit from model diversity. In practice, LSTM and GRU both collapse to repetitive high-frequency residue predictions on longer sequences, so their candidates are correlated rather than complementary. NovoBench [9] showed that no single model dominates across all evaluation conditions; effective ensembles require models with structurally distinct failure modes, which LSTM and GRU do not provide.

Wastewater coverage. Six identifications at 5% FDR is a modest yield for four mzML files. Our model was trained exclusively on clean Orbitrap E. coli spectra, and wastewater acquisitions differ substantially in noise profile, charge distribution, and peptide composition. Spectral noise augmentation during training partially mitigates this mismatch, but bridging it fully would likely require wastewater-specific training data.

References

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